OBSTETRICS

von Willebrand disease and pregnancy: a practical approach for the diagnosis and treatment

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von Willebrand disease is caused by either a quantitative or qualitative defect in von Willebrand factor (VWF). Patients may have extensive mucosal bleeding (because of platelet dysfunction) and prolonged bleeding after surgery (because of factor VIII deficiency). Up to 6 different subtypes of the disease have been described, and diagnosis is based on clinical suspicion and laboratory confirmation. Accurate diagnosis is of paramount importance because therapy will vary according to the subtype. Bleeding complications during pregnancy are more frequent when levels of the von Willebrand ristocetin cofactor assay and factor VIII levels are <50 IU/dL. In such cases, therapy before any invasive procedure or delivery must be instituted. The mainstays of therapy are desmopressin and plasma concentrates that contain von Willebrand factor. Delayed postpartum hemorrhage may occur, despite adequate prophylaxis. Frequent monitoring and continued prophylaxis and/or treatment are recommended for at least 2 weeks after delivery.

Key words: hemorrhage, pregnancy, von Willebrand disease

A 26-year-old patient was seen for routine prenatal care. She had a history of delayed postpartum hemorrhage 4 years earlier after an uncomplicated vaginal delivery. During a recent visit to the dentist, she experienced increased bleeding after a tooth extraction. Bleeding diathesis was considered, and the question is how to evaluate and treat this patient during the current pregnancy.

In 1926, Erik von Willebrand first described a familial bleeding disorder in which symptoms were more severe in children and young women. He also noted that the condition had autosomal inheritance and that blood transfusions improved bleeding symptoms. Since then, knowledge of this disease has grown exponentially and is currently recognized as the most common inherited bleeding disorder.[1-5]

PATHOGENESIS

Now known as von Willebrand disease (VWD), this disorder is caused by either a quantitative or qualitative defect in von Willebrand factor (VWF). The prevalence of VWD is as high as 1-2% in the general population.[6,7]

VWF, a large multimeric glycoprotein, is encoded by a gene that is located in the short arm of chromosome 12 and is synthesized and released from the endothelium and megakaryocytes.[8] As a result of vascular injury, VWF adheres to the subendothelial matrix that is exposed; with shear stress, the multimeric VWF uncoils, which leads to platelet adhesion, activation, and aggregation.

Besides the predominant role in platelet-induced hemostasis, VWF also binds factor VIII (FVIII) in plasma, thereby preventing its degradation and clearance. The latter action explains the reason that VWD is characterized by both extensive mucosal bleeding (because of platelet dysfunction) and prolonged bleeding after surgery (because of FVIII deficiency).

Interestingly, the disease is disproportionately symptomatic in women of child-bearing age and appears in the form of menorrhagias and/or postpartum hemorrhage. Numerous case series of pregnancies that have been complicated with VWD have been reported in the literature.[9-11] Overall, such women are at a higher risk of bleeding that is associated with invasive procedures, delivery, and both immediate and delayed postpartum hemorrhage.

Consequently, it is of utmost importance that the obstetrician is familiar with this condition.

The classification of VWD was updated in 2006 (Table 1) in an attempt to correlate disease subtype and optimal therapy.[12]

DIAGNOSIS

Suspicion for the presence of VWD usually arises from a patient's personal history of excessive bleeding and/or a similar family history. Importantly, a family history is frequently not present because family members may be affected by the

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TABLE 1
Classification of von Willebrand disease

Type	Explanation
1	Type 1 VWD is the classic and most common form that accounts for nearly 80% of all cases. This subtype is characterized by a partial quantitative deficiency in VWF. Inheritance is autosomal dominant, and the bleeding propensity is mild to moderate.
	The laboratory evaluation of this disease will typically show a decrease in plasma concentration of both VWF protein (von Willebrand factor antigen [VWF:Ag]) and factor VIII (FVIII). Qualitative functional assays of VWF function (von Willebrand factor ristocetin cofactor activity [VWF:RCo]) show a decrease in function that parallels the quantitative deficit of the protein.[13]
2	Type 2 VWD is mainly a qualitative/functional problem of the VWF (plasma concentrations of VWF are normal or near normal). Four variants of type 2 VWD have been described.
	Type 2A VWD is a variant in which, despite normal (or mildly reduced) plasma concentrations of VWF:Ag and FVIII, VWF-dependent platelet aggregation is compromised. This is demonstrated by a significantly decreased VWF functional test (VWF:RCo). The main problem here is a deficit in the proportion of large VWF multimers. These large multimers are the ones that ultimately interact with platelets to achieve aggregation. The condition may be inherited in either autosomal dominant or recessive fashion.
	Type 2B VWD is due to a pathologic increase in binding between platelets (through glycoprotein Ib) and VWF. Platelets bind to the mutant large multimers, and these complexes are sequestered in the microcirculation and subsequently dissolved by the metalloproteinase ADAMTS 13. This then leads to a decrease in large multimers and thrombocytopenia. Pregnancy and surgery frequently aggravate thrombocytopenia in these patients.[14] Laboratory diagnosis rests on demonstration of a pathologically increased ristocetin-induced platelet aggregation. Inheritance is autosomal dominant.
	Type 2M VWD is characterized by decreased platelet adhesion to VWF due to minimal affinity between VWF and the platelet surface glycoprotein Ib. Unlike types 2A and 2B, the concentration of large VWF multimers is normal in this subtype. The diagnosis rests on the demonstration of poor platelet aggregation in vitro (VWF:RCo) in the presence of a normal amount of large VWF multimers on gel electrophoresis.[13] Inheritance of this subtype may be autosomal dominant or recessive.
	Type 2N VWD is caused by minimal binding between VWF and FVIII. Quantitative (VWF:Ag) and qualitative (VWF:RCo) testing for VWF are normal; however, the FVIII level is low. The main differential diagnosis in this situation is hemophilia A. Inheritance is autosomal recessive.
Type 3	Type 3 VWD is rare. The prevalence has been estimated at 1 per 1,000,000. This is a severe quantitative defect in VWF that is characterized by undetectable levels of VWF:Ag in plasma. Consequently, FVIII levels will also be very low. Bleeding propensity in this subgroup is obviously very high. Inheritance is autosomal recessive.
Acquired	Acquired VWD has also been described. Autoimmune diseases may lead to increased clearance or inhibition of VWF (mediated by autoantibodies that cross-react with VWF). Certain conditions (such as ventricular septal defects, aortic stenosis, and pulmonary hypertension) will increase fluid shear stress and may lead to increased proteolysis of VWF by ADAMTS 13. Certain medications (hydroxyethylstarch, valproic acid, ciprofloxacin) may induce acquired VWD.[13]

VWD, von Willebrand disease; *VWF*, von Willebrand factor.

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milder forms of the disease and have minimal-to-no symptoms.

Symptoms usually involve mild-to-moderate mucocutaneous bleeds, such as epistaxis, menorrhagia, bleeding after dental extractions, ecchymoses, gingival bleeding, and prolonged bleeding from minor cuts. Patients with menorrhagia and no documented pelvic anomalies should undergo testing for VWD. Postoperative and delayed postpartum hemorrhage may also be present. Most of these events do not require blood transfusions or even doctor visits.[13]

Patients with the rare type 3 disease may have life-threatening bleeding episodes, such as intracranial and gastrointestinal hemorrhages.

Once the clinical suspicion is present, the next step is laboratory confirmation. At the University of Texas in Galveston, we use the following protocol. We start with a basic hemostasis evaluation that includes a complete blood cell count, platelet count, activated partial thromboplastin time (aPTT), prothrombin time, and fibrinogen levels. The initial workup is mainly to rule out other potential causes of bleeding (eg, severe thrombocytopenia) and not particularly directed specifically at the diagnosis of VWD. The aPTT is often normal in VWD, and only the most severe deficiencies with very low levels of FVIII will have a prolonged aPTT.[13] If on initial evaluation the aPTT is prolonged, it should correct on a 1:1 mixing with normal plasma to be consistent with VWF (if it does not correct, the presence of a FVIII inhibitor should be suspected).

If the initial test results return normal or show a prolonged aPTT that corrects with mixing, further testing is indicated. The following 3 tests should follow. The first test is the VWF protein (VWF:Ag) assay, which is an immunoassay that measures the VWF protein in plasma. The second test to be performed is a functional assay (VWF:RCo) that measures the interaction between VWF and platelets. The binding between these 2 and subsequent formation of platelet clumps is stimulated by the addition of the antibiotic ristocetin. Platelets bind mainly to the large VWF multimers through their surface glycoprotein 1b.The third test, the FVIII assay, measures the activity of FVIII, which is a surrogate for the activity of VWF as a carrier protein for FVIII.

TABLE 2
Common laboratory findings in von Willebrand disease

Subtype[a]	von Willebrand factor antigen	von Willebrand factor ristocetin cofactor activity	von Willebrand factor ristocetin cofactor activity/ von Willebrand factor antigen	Factor VIII	Low dose ristocetin-induced platelet aggregation	Multimer assay
Type 1	Low	Low	>0.5-0.7	Low or normal	No reaction	Normal
Type 2A	Low	Low	<0.5-0.7	Low or normal	No reaction	Decrease in large multimers
Type 2B	Low	Low	<0.5-0.7	Low or normal	Positive	Decrease in large multimers
Type 2M	Low	Low	<0.5-0.7	Low or normal	No reaction	Normal
Type 2N	Normal to low	Normal to low	>0.5-0.7	Low	No reaction	Normal
Type 3	Absent	Absent		Low	No reaction	Absent

[a] See Table 1 for explanations of each subtype.

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Other specialized tests may follow to further classify the disease. Gel electrophoresis may provide information on the abundance of different-sized VWF multimers, which would aid in the diagnosis of subtypes 2A and 2B, where large multimers are decreased.

A binding assay is available to evaluate interaction between VWF and FVIII. Minimal interaction will point towards type 2N VWD.[15]

The low-dose ristocetin-induced platelet aggregation test is valuable in diagnosing type 2B VWD. In this test, low doses of the antibiotic ristocetin are added to the patient's sample. In type 2B disease, because of the high affinity of VWF and platelets, even such low concentrations of ristocetin will induce platelet aggregation. This test will be negative in all other subtypes of the disease.

Reference laboratories frequently will perform a ratio to compare the amount of qualitative-to-quantitative dysfunction present (VWF:RCo/VWF:Ag). This will be useful in the differentiation of type 1 and type 2 VWD variants. If the ratio is <0.5-0.7, the patient is most likely affected with type 2 disease, because such a low ratio is consistent with dysfunctional VWF (a higher ratio is consistent with type 1 disease because the VWF:Ag will be lower).

Both the VWF:Ag and VWF:RCo assays are reported in international units per deciliters. A value <30 IU/dL is diagnostic for VWD; however, some patients with VWD may have values between 30 and 50 IU/dL.[13] This subgroup of patients should be treated in the same fashion as other patients with VWD before any invasive procedure during pregnancy.

Occasionally, repeated testing for VWD is needed because conditions such as pregnancy, stress, surgery, and inflammation will increase levels of VWF and mask lower baseline values.

Table 2 shows the typical laboratory findings according to the various VWD subtypes.

Genetic Considerations

The VWF gene is located near the tip of the short arm of chromosome 12. Mutations that cause VWD have been identified throughout this gene. Such mutations include large deletions, insertions, splice-site mutations, and nonsense and missence mutations. The use of gene-specific polymerase chain reaction primers has facilitated the diagnosis in some subtypes of the disease. A database of such mutations is available online at the University of Sheffield.[16]

Prenatal genetic testing may be achieved by either chorionic villous sampling or amniocentesis. When testing is performed for VWD, knowing the subtype is imperative. Regarding type 1 VWD, the genotype-phenotype correlation can be inconsistent. The phenotype is relatively mild and very responsive to current medical management. As a result of the risks that are associated with testing, current recommendations are that prenatal testing is of limited use in this population.[8] Further clarification of the molecular basis of type 1 disease is required before clinical genetic testing is implemented.

In the case of type 2 VWD, the correlation between phenotype and genotype is more consistent. This consistent correlation, coupled with the fact that mutations are relatively well localized, provides a good rationale for genetic testing.[17] Prenatal diagnosis is certainly important for women who are known to be carriers of the most severe type 3 disease.[18] The most common mutation reported in type 3 VWD is a frameshift mutation in exon 18.[17]

Treatment

Management of VWD during pregnancy should ideally include obstetricians, hematologists, and blood bank personnel.

During the course of pregnancy, 1 of the many physiologic changes is an increase in several components of the clotting cascade, which include both FVIII and VWF levels. Consequently, many patients with VWD reach normal levels of both VWF and FVIII at term. The lat-

TABLE 3
Clinical pearls for the management of von Willebrand disease during pregnancy

A. Consider diagnosis in patients with mild-to-moderate mucocutaneous bleeds, such as epistaxis, menorrhagia, delayed postpartum hemorrhage, bleeding after dental extractions, ecchymoses, gingival bleeding, and prolonged bleeding from minor cuts. Patients with menorrhagia and no documented pelvic anomalies should be screened for von Willebrand disease.
B. Detailed laboratory testing is basic to identify correctly the subtype of the disease. Treatment is tailored according to subtype.
C. Once diagnosed, follow-up factor VIII levels should be monitored at least once every trimester.
D. Offer genetic counseling.
E. Maintain factor VIII levels at >50 IU/dL before any invasive procedures, at the time of delivery, and during the postpartum period. Ideally, maintain such levels for up to 2 weeks after delivery.
F. If possible, avoid procedures such as pudendal blocks, episiotomies, and operative vaginal deliveries.

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ter explains the low incidence of bleeding complications antenatally.

There seems to be no consensus in the literature as to how often levels of FVIII and VWF:RCo must be monitored. However, there is agreement that, at the very minimum, levels should be determined during the third trimester.[1,6] At the University of Texas, we monitor FVIII and VWF:RCo levels once every trimester. In addition, these tests should be ordered if there is anticipated risk of bleeding at any point in the pregnancy or before any invasive procedure that would increase the likelihood of bleeding (such as chorionic villus sampling, amniocentesis, or cerclage).[1] Before amniocentesis or chorionic villus sampling, we recommend a level of FVIII and VWF:RCo of ≥50 IU/dL. Treatment options are discussed later.

Pregnant women with VWD are at higher risk of hemorrhagic complications, mainly during the postpartum period.[19] Such events are far more common if both FVIII and VWF:RCo levels are <50 IU/dL. Overall, 16-29% of women with VWD will have postpartum hemorrhage within the first 24 hours of delivery, and 20-29% of women will experience delayed postpartum bleeding.[20] The third stage of labor should be managed actively. Procedures that potentially could increase the risk of hemorrhage (eg, pudendal blocks, episiotomies, fetal scalp electrodes, or operative vaginal deliveries) should be avoided if possible.

Vaginal delivery is considered safe for women with VWD types1 and 2. In patients with type 3 disease, a prolonged second stage of labor should be avoided, and early cesarean delivery may be considered in such cases.[21] The occurrence of central nervous system bleedings in neonates with VWD (including the most severe form, type 3) is extremely rare.[22] In most cases, route of delivery should be dictated by obstetric indications.

FVIII level is the best predictor of bleeding complications. It is recommended that VWF:RCo and FVIII levels be checked in the third trimester close to delivery. Levels of ≥50 IU/dL are considered safe. In fact, if such values are present and the coagulation screen is normal, regional anesthesia may be considered safe.[11] Values of >50 IU/dL should be maintained for at least 3-5 days after delivery.

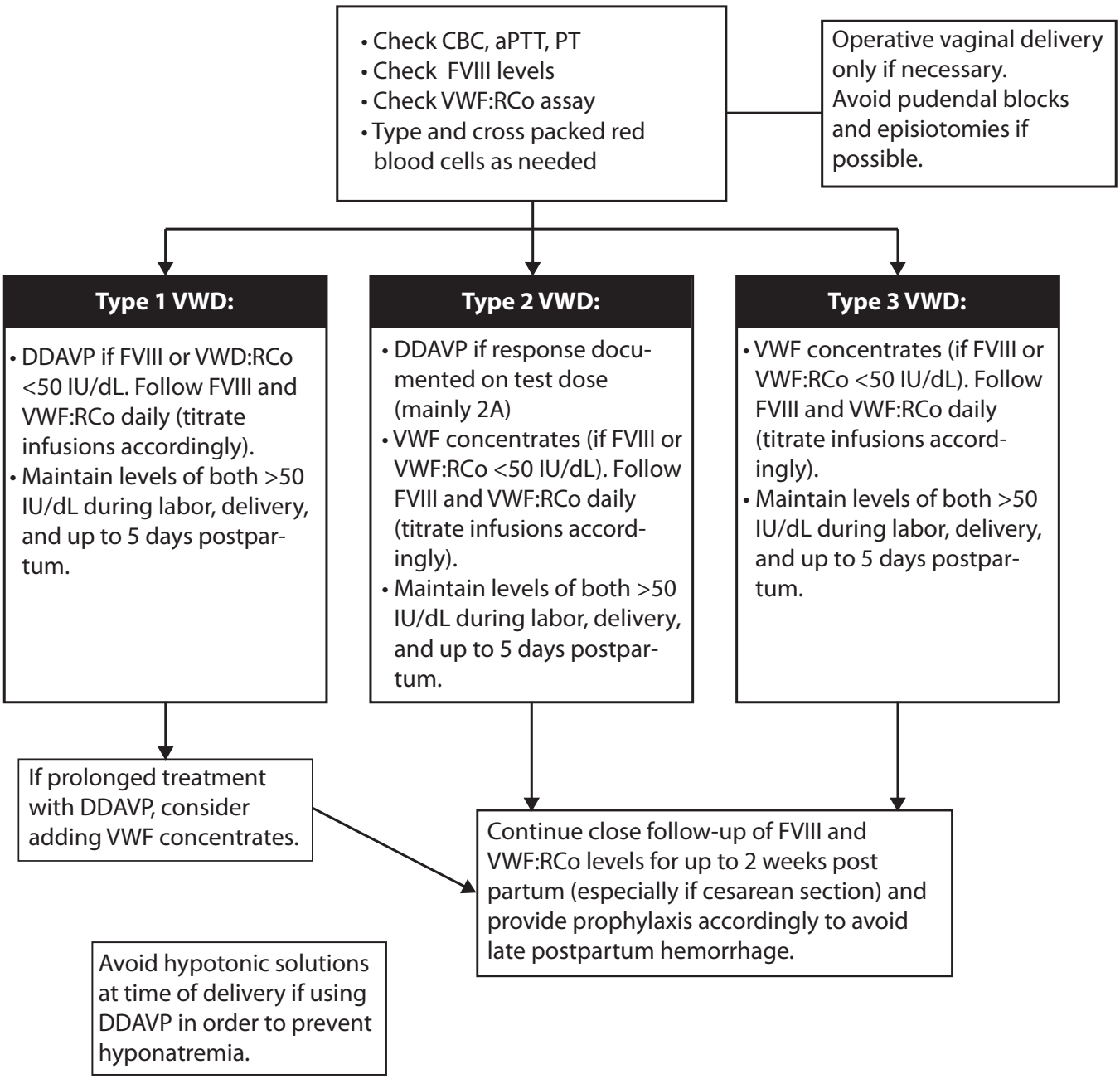
Pregnancy-induced elevation of VWF and FVIII levels is obviously a transient event, and levels will return to baseline within 7-21 days. This drop explains the high incidence of delayed postpartum hemorrhage. Frequent observation of these patients in the first weeks after delivery is of paramount importance. On discharge, patients with VWD should be instructed to report any excessive bleeding and to avoid nonsteroidal antiinflammatory medications, because they may alter platelet function and increase bleeding complications.[1] Ultimately, delayed postpartum hemorrhage is a very real possibility as coagulation factors return to baseline and can occur despite adequate prophylaxis. Therefore, frequent monitoring and continued prophylaxis and/or treatment is recommended for at least 2 weeks after delivery.

Patients with FVIII levels of <50 IU/dL should be treated before delivery and whenever bleeding is anticipated. The mainstays of therapy are desmopressin (DDAVP; 1-deamino-8-D-arginine vasopressin) and plasma concentrates that contain VWF. Fibrinolysis inhibitors are considered adjuvant therapy.

DDAVP is a synthetic derivative of the antidiuretic hormone vasopressin. It acts mainly on V2 receptors that are located in the renal collecting ducts that favor water retention. V2 receptors are also located in the endothelial cells where DDAVP will induce release of preformed VWF. The typical dose is 0.3 μg/kg intravenously (maximum dose, 25-30 μg) given over 30 minutes. The observed peak increment in FVIII and VWF levels usually occurs 30-90 minutes after the infusion. Alternatively, DDAVP may be administered intranasally at a dose of 300 μg. Intravenous administration is the preferred route for prophylaxis of surgical bleeding.[13]

Because the elevation of factors lasts for only 8-10 hours after administration, DDAVP should be administered every 12-24 hours. The main problem with this approach is tachyphylaxis (decreased responses with repeated use). The latter may be avoided by monitoring FVIII response after each dose and tailoring repeated treatments accordingly. In general, if treatment is required for >3 days, VWF concentrates should be

FIGURE
Peripartum management of von Willebrand disease



CBC indicates complete blood count; aPTT, activated partial thromboplastin time; PT, prothrombin time; VWF:RCo, von Willebrand factor ristocetin cofactor assay; FVIII, factor VIII; DDAVP, desmopressin.

aPTT, activated partial thromboplastin time; *CBC*, complete blood count; *DDAVP*, desmopressin; *FVIII*, factor VIII; *PT*, prothrombin time; *VWF:RCo*, von Willebrand factor ristocetin cofactor assay.

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used as a supplement to therapy with DDAVP.[23]

Patients with type 1 disease respond to DDAVP, but patients with type 3 do not. In type 2 disease, the main problem is that, despite an increase in secretion of VWF after DDAVP, the VWF secreted will retain its intrinsic molecular dysfunction. Consequently, the preferred therapy for type 2 disease is the use of VWF concentrates.[6] However, a small subset of patients with type 2 VWD re-

spond to desmopressin. Identification of those individuals requires a test dose of DDAVP and subsequent measurement of VWF:RCo 4 hours after the dose. If the VWF:RCo corrects after the dose, DDAVP is an acceptable treatment for those particular patients. In the past, type 2B disease was considered to be a contraindication for DDAVP use because the secretion of multimers with high affinity for platelets lead to thrombocytopenia. However, such an effect has been noted to be transient and not associated with bleeding or thrombosis.[24]

Recent data indicate that some individuals with type 1 disease have accelerated clearance of VWF; therefore, even patients with type 1 disease may benefit from a test dose of DDAVP and subsequent measurement of VWF:RCo to document treatment efficacy.

Minor side-effects of DDAVP include flushing, headache, gastrointestinal complaints, and transient hypo- or hypertension. Repeated dosing may lead to water retention and hyponatremia. It is of paramount importance to limit fluid intake and to follow sodium plasma levels during prolonged therapy.[25] Myocardial infarction has been reported in elderly patients with coronary artery disease after the use of DDAVP.[26] In general, the drug should not be used in people with unstable coronary artery disease because the large VWF multimers that are secreted will aggregate platelets and could lead to myocardial infarction.[6]

DDAVP is safe for the fetus because it does not cross the placenta in detectable amounts and has little-to-no oxytocic effect.[6] DDAVP is a pregnancy category B medicine.

VWF concentrates include Humate-P and Alphanate SD/HT. These are plasma-derived concentrates of VWF and FVIII that are available in the United States for treatment of patients with VWD who are not candidates for DDAVP treatment. Adverse reactions to Humate-P and Alphanate are rare but may include rash, pruritus, and urticaria. When used for several days, one should follow levels of VWF:RCo (do not exceed 200 IU/dL) and FVIII (do not exceed 250-300 IU/dL) because high levels of FVIII have been associated with venous thromboembolism.[27] Thromboembolic prophylaxis in patients who undergo surgery with VWD should be used in the same fashion as in patients without the disease.[6]

The usual loading dose for these products is 40-60 IU/kg.[12] VWF concentrates are dosed primarily on the basis of VWF:RCo units and secondarily on the basis of FVIII units. Infusions ideally should start daily before delivery until levels of 50 IU/dL are achieved for both VWF:RCo and FVIII. Such levels should be maintained for at least 3-5 days after delivery. Levels should be followed daily, and the need for repeated doses should be tailored accordingly. Maintenance doses usually range between 20 and 40 IU/kg every 12-48 hours. As stated before, because of the risk of delayed postpartum hemorrhage, laboratory monitoring may be needed for ≥2 weeks after delivery to correctly identify those patients who will require prolonged prophylaxis with either VWF concentrates or DDAVP.

In acute bleeding cases, hemostasis may be achieved in VWD with 10 bags of cryoprecipitate. However, this blood product does not undergo viral inactivation, which poses a significant risk of disease transmission. Fresh frozen plasma may also be used to correct VWD; however, large amounts of volume will be needed to achieve hemostasis.

Platelet concentrates contain 15% of total blood VWF; consequently, platelet transfusions are considered to be an adjunctive therapy in cases in which bleeding is not controlled, despite adequate FVIII levels.[28]

Antifibrinolytics are considered second-line therapy for patients with VWD. They are mostly used to treat mild mucocutaneous bleeding in VWD because mucosal tracts typically have rich fibrinolytic activity.[6] Aminocaproic acid is administered as a loading dose of 5 g, starting 1 hour before an invasive procedures followed by a drip of 1 g/hr until bleeding is controlled. The maximum dose for aminocaproic acid is 24 g in 24 hours. Side-effects include nausea, vomiting, and, rarely, thrombotic complications. Antifibrinolytics should be avoided in patients with gross hematuria, because renovascular thrombi may fail to lyse and cause urinary tract obstructions.[6]

An algorithm for the treatment of VWD in the peripartum period is shown in the Figure.

COMMENT

The patient in the vignette at the beginning of the article required laboratory evaluation for VWD that included a basic coagulation panel, VWF;Ag assay, VWF:RCo assay, and FVIII levels. If consistent with VWD, levels of FVIII and VWF:RCo should be obtained in each trimester, and treatment should be instituted if the levels are <50 IU/dL before any invasive procedure and delivery. Close observation for at least 2 weeks after delivery is mandatory.

Despite the therapeutic physiologic effect of pregnancy on patients with VWD, bleeding complications remain a significant risk to both mother and fetus. Because the risk of bleeding varies by subtype, accurate diagnosis of the disease is an essential step of care to provide recommendations regarding optimal therapy and genetic counseling. Overall, the preferred management for pregnancy that is complicated by VWD is anticipating complications by monitoring bleeding parameters. Therefore, with adequate monitoring, prophylaxis, and observation, patients with VWD can be expected to tolerate the course of pregnancy with minimal risk. Clinical pearls for the management of VWD in pregnancy are listed on Table 3. ■

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